

Group 10
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Fall 22/23



Feliciano School of Business

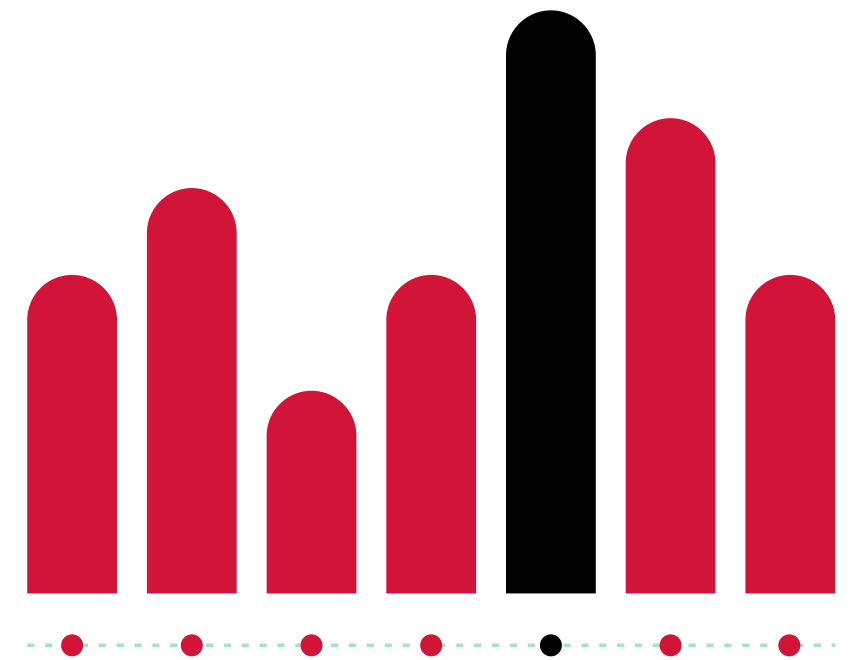
Nutrition Study

INFO 589 Applied Statistics for Business Analytics

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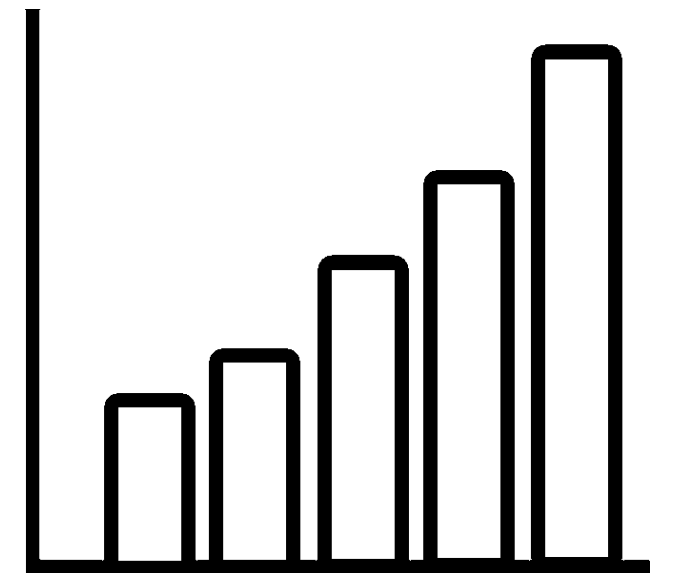
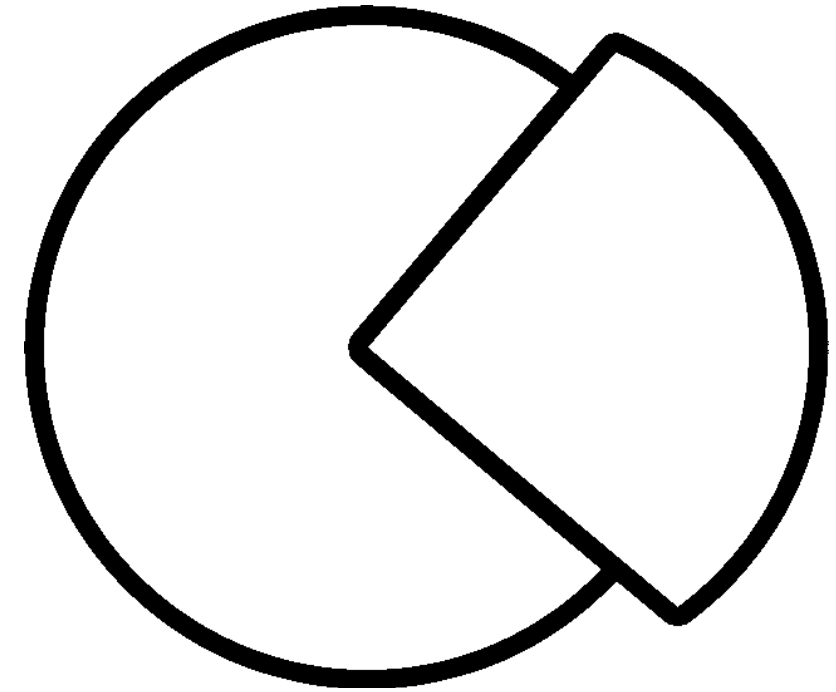
Objectives

- To test how calories, age, fat, fiber, alcohol, cholesterol, smoking, and vitamin use affect body wellness.
- To test if they vary in terms of gender, age groups, and habits.
- To test if cholesterol can be a significant factor that responds to all other variables that will determine the body's well-being.
- To test the overall relationships among the variables.



Applied Tests

- T-test
- Chi-Square Test
- ANOVA
- Multivariate Regression Analysis
- Binary Logistic Regression Analysis

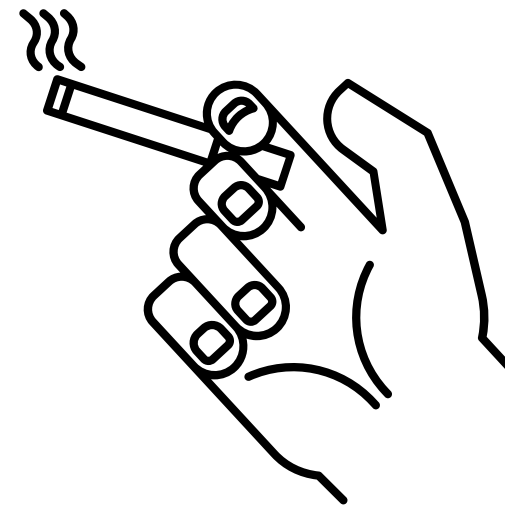


DATA

Qualitative Data



Female and Male - Binary
(0=Female, 1=Male)
42 Males / 273 Females



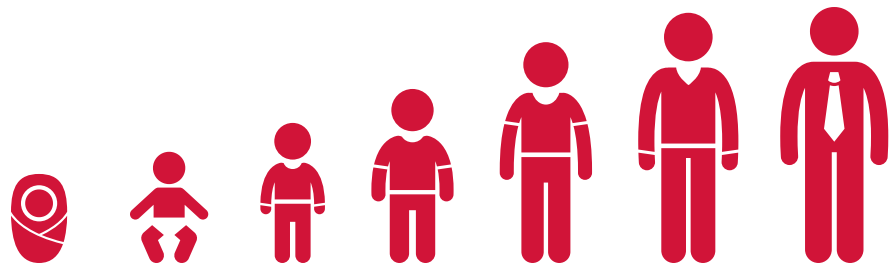
Females | 36 Smokes | 237 Non-Smokers
Males | 7 Smokers | 34 Non-Smokers



Vitamins Use
Regular, Occasional, No

DATA

Quantitative Data



Age measured in years

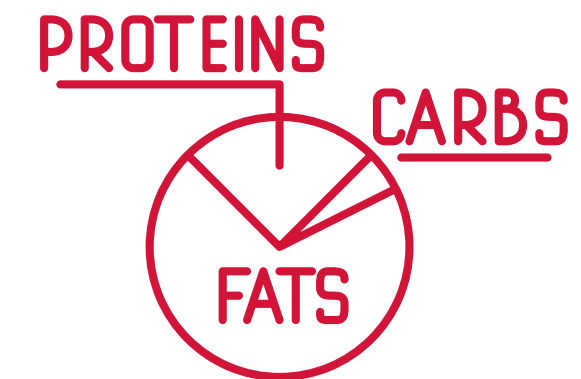
Categorical Subcategory:

Under 35, Between 35-50, Over50.



Measured in kilocalories

Daily calories consumption



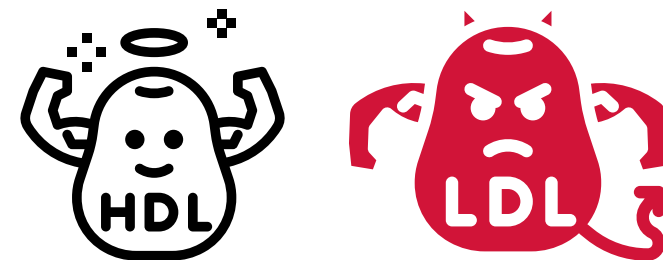
Measured in milligrams

Fat Level



Measured in Grams

Fiber Level



Measured in milligrams

Cholesterol Level



Measured by percentage.

Subcategory:

Alcohol consumption Yes or No - Binary (0=No, 1=Yes)

Decision Rule

Hypothesis & Level of significance & P-value

Hypothesis: there are two main types of hypotheses:

- H_0 : The null hypothesis: It is a statement about the population that either is believed to be true or is used to put forth an argument unless it can be shown to be incorrect beyond a reasonable doubt.
- H_a : The alternative hypothesis: It is a claim about the population that is contradictory to H_0 and what we conclude when we reject H_0 .

Level of significance & P-value

The significance level, or alpha (α), is a value the researcher sets as the threshold for statistical significance in advance. It is the maximum risk of making a false positive conclusion (Type I error) that you are willing to accept. The p-value is compared to the significance level in a hypothesis test to decide whether to reject the null hypothesis.

-If the p-value is higher than the significance level, the null hypothesis is not refuted, and the results are not statistically significant.

-If the p-value is lower than the significance level, the results are interpreted as refuting the null hypothesis and reported as statistically significant.

Testing | t-test

Area of interest

We used a t-test to determine if there is a significant difference between the means of the two groups (Male and Female) in relation to cholesterol, fiber, age, fat, smoking, vitamin use, and calories.

Assumptions:

1. Male and female populations are normally distributed.
2. The variances of the two groups are assumed unequal.
3. Data is obtained as a random sample from populations.
4. The sample is large enough, with more than 30 observations for both populations.

Hypothesis:

H0: $\mu_{\text{male}} - \mu_{\text{female}} = d_0$

HA: $\mu_{\text{male}} - \mu_{\text{female}} \neq d_0$

Level of Significance:

5% level of significance

Calories				
t-Test: Two-Sample Assuming Unequal Variances				
	Variable 1	Variable 2		
Mean	2155.786	1741.404		
Variance	840091.9	384734.9		
Observations	42	273		
Hypothesized Mean Difference	0			
df	47			
t Stat	2.831898			
P(T<=t) one-tail	0.003396			
t Critical one-tail	1.677927			
P(T<=t) two-tail	0.006792			
t Critical two-tail	2.011741			

Results

Variable	P. Value	Conclusion
Age	1.73637E-06	Null hypothesis is rejected
Fat	0.000959362	Null hypothesis is rejected
Fiber	0.379798465	Null hypothesis is not rejected
Alcohol	0.097555954	Null hypothesis is not rejected
Cholesterol	0.000116767	Null hypothesis is rejected
Smoking	0.575257764	Null hypothesis is not rejected
Vitamin	0.667125462	Null hypothesis is not rejected
* Assuming Regular /No		

Conclusion: At the 5% significance level, we reject H0 for Calories, Age, Fat, and Cholesterol since the P-value of the two-tailed test is less than 0.05 in all of them. Therefore, We conclude that the average between males and females differs in terms of those variables. On the other hand, the p-values of Fiber, Alcohol, Smoking, and Vitamins are more greater than 0.05. Therefore, we can't reject H0, which indicates that the average in males and females in terms of those variables don't differ.

Smokers & Non Smokers

We used a t-test to determine if there is a significant difference between the means of the two groups (Smokers and Non Smokers) in relation to cholesterol, fiber, age, fat, gender, vitamin use, and calories.

Results

Variable	P. Value	Conclusion
Calories	0.332496069	Null hypothesis is not rejected
Age	0.005241611	Null hypothesis is rejected
Fat	0.136734008	Null hypothesis is not rejected
Fiber	0.001508887	Null hypothesis is rejected
Alcohol	0.29500293	Null hypothesis is not rejected
Cholesterol	0.14583502	Null hypothesis is not rejected
Gender	0.575140254	Null hypothesis is not rejected
Vitamin	0.022070653	Null hypothesis is rejected

Hypothesis:

H0: $\mu_{\text{non smokers}} - \mu_{\text{smokers}} = 0$

HA: $\mu_{\text{non smokers}} - \mu_{\text{smokers}} \neq 0$

Level of Significance:

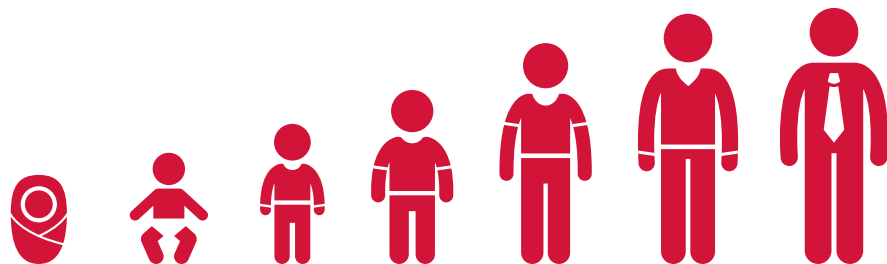
5% level of significance

Conclusion: We reject the null hypothesis in age, fiber, and vitamin since the p-value is less than the chosen alpha. Therefore, at a 5% significance level, we conclude that the average age, fiber level, and vitamin use between smokers and non-smokers are significantly different. On the other hand, we see that the means do not differ between smokers and non-smokers with calories, fat, alcohol, cholesterol, and gender.

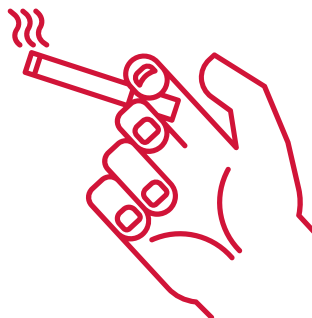
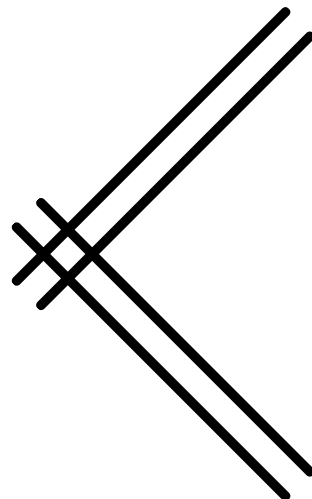
Testing | Chi Square Test

Area of interest

We wanted to determine whether there is a statistically significant difference between the categorical variables Age & Smoking, Age & Alcohol, and Age & Vitamin Use placed in a contingency table that tests whether they are dependent or independent of each other.



Age measured in years
Categorical Subcategory:
Under 35, Between 35-50, Over50.



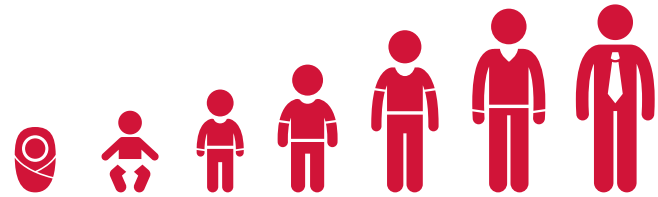
Females | 36 Smokes | 237 Non-Smokers
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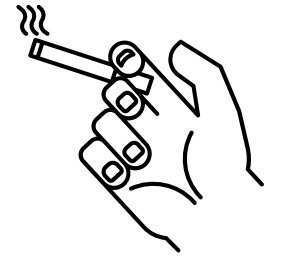
Measured by percentage.
Subcategory:
Alcohol consumption Yes or No - Binary (0=No, 1=Yes)



Vitamins Use
Regular, Occasional, No



Age & Smoking



Hypothesis:

H_0 = Age group and Smoking are independent. 5% level of significance

H_A = Age group and Smoking are dependent.

Level of significance:

Frequency Table	Under 35	Between 35 - 50	Over 50	Grand Total
No Smoke	121	32	119	272
Smoke	11	11	21	43
Grand Total	132	43	140	315

Expected Frequency	Under 35	Between 35 - 50	Over 50	$O_i - E_i$	$O_j - E_j$	$O_k - E_k$
No Smoke	114	37	121	0	1	0
Smoke	18	6	19	4	2	0
Grand Total	132	43	140	5	3	0

$df = (r-1)(c-1)$	2
$\chi^2_{df=2}$	8.30091634
$P(\chi^2 \geq 9)$	0.0157572



Conclusion: we found the p-value = 0.015.

Therefore, at the 5% significance level, we can conclude that we reject the null hypothesis. For instance, Age group and Smoking habits are dependent; people under 30 and over 50 are less likely to smoke than those between 35-50 due to the proportion of people that smoke being 26% (11/43) which is higher than 8% (11/132) and 7% (21/140) according to our data.



Hypothesis:

Level of significance:

5% level of significance

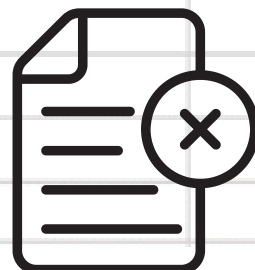
Conclusion: Since P-value = 0.002262, less than 0.05 at the 5% significance level we conclude that Age group and consuming alcohol are dependent according to our data.



Level of significance:

5% level of significance

Conclusion: at 5% we reject our H0, concluding that Age group and Vitamin use are dependent. Additionally, people between 30-50 and over 50 are more likely to use vitamins than under 35.



Testing | ANOVA Test One -Way ANOVA Test

Area of interest

To determine whether there are any statistically significant differences between the means of calories, fat, fiber, and cholesterol in terms of Age.

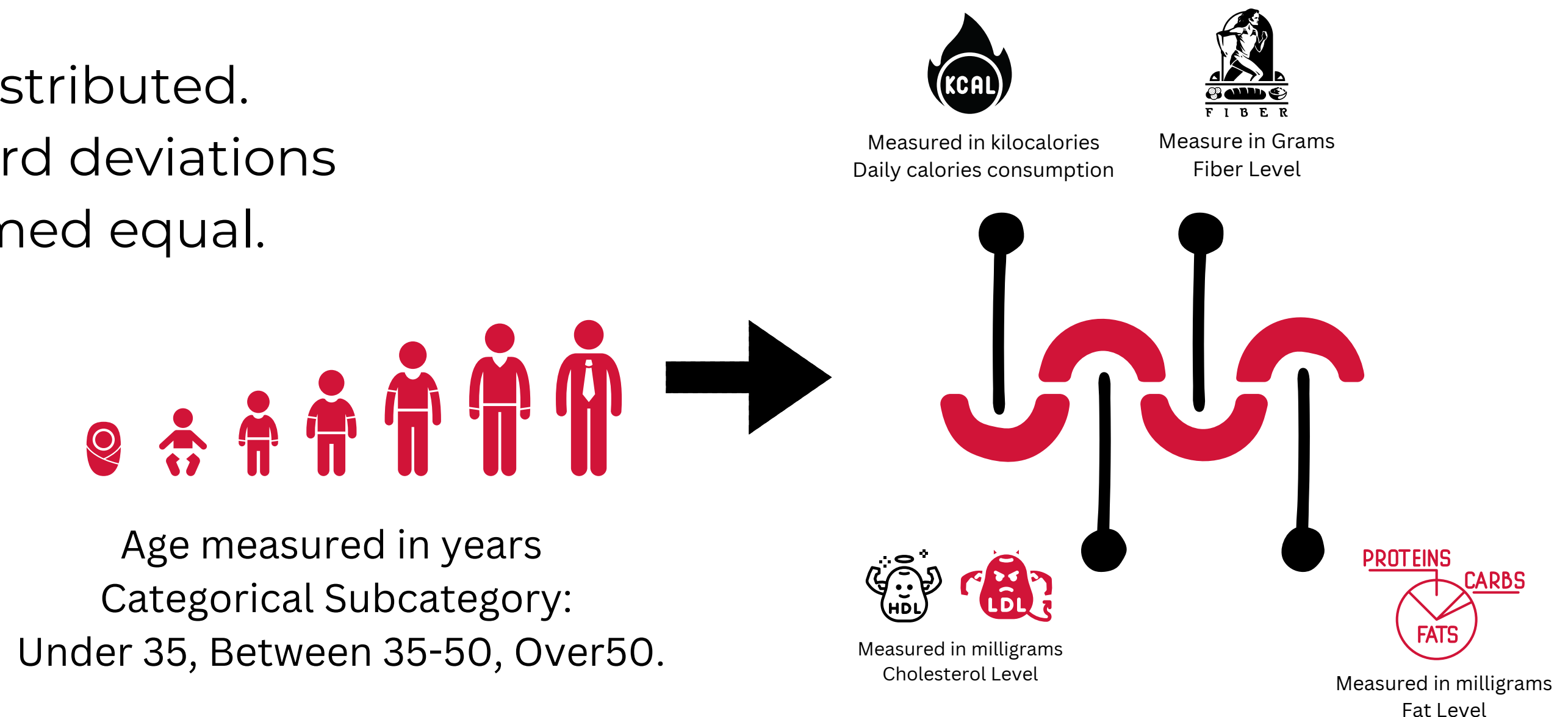
Assumptions

- The data is normally distributed.
- The population standard deviations are unknown but assumed equal.
- Samples are selected independently.

Hypothesis

$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$

H_a : Not all population means are equal



Test 1: Calories Intake vs Age Group

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
Under 35	43	89979.3	2092.54186	530602.281		
Between 35 - 50	140	249921.5	1785.15357	373804.617		
Over 50	132	226045.4	1712.46515	506710.947		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	4718734.91	2	2359367.45	5.23471426	0.00580635	3.02468142
Within Groups	140623272	312	450715.614			
Total	145342006	314				

Test3: Fat Intake vs Age Group

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
Under 35	43	533.9	12.4162791	20.8609192		
Between 35 - 50	140	1787.4	12.7671429	30.1930134		
Over 50	132	1707.1	12.9325758	29.3075567		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	8.76147111	2	4.38073555	0.1533603	0.85788522	3.02468142
Within Groups	8912.27739	312	28.5649916			
Total	8921.03886	314				

Test 2: Fiber Intake vs Age Group

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
Under 35	43	3910.8	90.9488372	1413.67589		
Between 35 - 50	140	10650.8	76.0771429	1048.74839		
Over 50	132	9703.9	73.5143939	1100.09055		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	10089.123	2	5044.56152	4.5063647	0.01176644	3.02468142
Within Groups	349262.277	312	1119.43037			
Total	359351.4	314				

Test 4: Cholesterol Level vs Age Group

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
Under 35	43	11683.7	271.713953	19706.9922		
Between 35 - 50	140	33973.5	242.667857	15071.8157		
Over 50	132	30717.9	232.711364	19071.8693		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	49349.9219	2	24674.961	1.42011782	0.24324335	3.02468142
Within Groups	5421090.93	312	17375.2914			
Total	5470440.85	314				

Analysis and inference for a one-way ANOVA test: P-Value

Calories & Age	Fiber Intake & Age	Fat Intake & Age	Cholesterol Level vs Age
0.00580635	0.01176644	0.85788522	0.24324335

At the 5% significance level we can conclude that the amount of calories and fat level by age group differs from each other. On the other hand, the mean fat and mean cholesterol level do not differ from each other, we may say that the mean of these variables may assume equal.

Test 1: Calories Intake vs Age Group

- H₀: All age groups consume the same amount of calories.
- H_A: At least one of the age groups consume a different amount of calories

Test 2: Fiber Intake vs Age Group

- H₀: All age groups have the same fat level.
- H_A: At least one of the age groups has different fat levels.

Test3: Fat Intake vs Age Group

- H₀: All age groups have the same fiber level
- H_A: At least one of the age groups has different fiber levels.

Test 4: Cholesterol Level vs Age Group

- H₀: All age groups have the same cholesterol level.
- H_A: At least one of the age groups has different cholesterol levels.

The Tukey's Method.

Area of interest

To test which groups differ with mean calories and fat levels

Contrast	Difference	Standardized difference	Critical value	Pr > Diff	Significant
Under 35 and Over 50	380.0767	3.2242	2.3550	0.0040	Yes
Under 35 and Btw 35-50	307.3883	2.6261	2.3550	0.0245	Yes
Btw 35 to 50 and Over50	72.6884	0.8924	2.3550	0.6455	No
Tukey's d critical value:			3.3304		

-Mean calorie intake differ between age group 1 and age group 2, as well as age group 1 and age group 3.

Contrast	Difference	Standardized difference	Critical value	Pr > Diff	Significant
Under 35 and Over 50	17.4344	2.9676	2.3550	0.0090	Yes
Under 35 and Btw 35-50	14.8717	2.5494	2.3550	0.0302	Yes
Btw 35 to 50 and Over50	2.5627	0.6314	2.3550	0.8030	No
Tukey's d critical value:			3.3304		

-Average fat levels differ between age group 1 and age group 2, as well as age group 1 and age group 3.

Testing | ANOVA Test Two -Way ANOVA Test

Area of interest

To test for interaction of cholesterol levels with smoking habits and age group.

Hypothesis

-H0: There is no significant interaction of cholesterol levels among smokers and nonsmokers and different age groups.

-HA: At least one of the groups differs.

Source	DF	Sum of squares	Mean squares	F	Pr > F
Between Groups	5	118062.3793	23612.4759	1.3632	0.2380
Within Groups	309	5352378.4726	17321.6132		
Total	314	5470440.8519			

Source	Value	Standard error	t	Pr > t	Lower bound (95%)	Upper bound (95%)
Intercept	305.4909	39.6824	7.6984	0.0000	227.4090	383.5728
Under 35	-36.2273	56.1194	-0.6455	0.5191	-146.6517	74.1972
Btw 35-50	-48.5100	48.9850	-0.9903	0.3228	-144.8963	47.8764
Over 50	0.0000	0.0000				
Smoker	-79.3959	41.4469	-1.9156	0.0563	-160.9497	2.1580
Non Smoker	0.0000	0.0000				
Under 35* Non Smoker	82.6885	61.9180	1.3355	0.1827	-39.1458	204.5227
Under 35 * Smoker	0.0000	0.0000				
Btw 35-50*Non Smoker	62.5569	51.8483	1.2065	0.2285	-39.4634	164.5773
Btw 35-50*Smoker	0.0000	0.0000				
Over 50*Non Smoker	0.0000	0.0000				
Over 50*Smoker	0.0000	0.0000				

Conclusion: The probability values indicate that cholesterol levels do not interact with smoking habits and any of the age groups.

Multivariate Regression Analysis

Area of interest

To estimate the effect of nutritional factors like Calories, Fat, Gender, Age, Alcohol, Smoker/Nonsmokers and Vitamin Use on Cholesterol

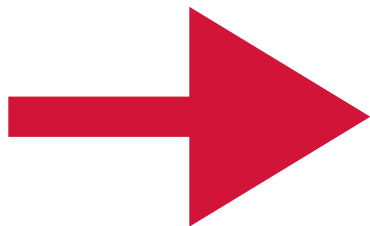
Hypothesis:

-H0: All coefficients in the model are equal to zero. None of the independent variables have a significant relationship with cholesterol.

-HA: At least one variable has an effect.

General guidelines to interpret test results

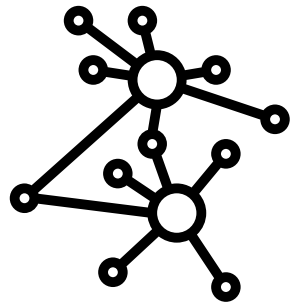
Cholesterol Level	
Below 200 mg/dL	Desirable
200-239 mg/dL	Borderline high
240 mg/dL and above	High



General Correlation between the factors

Bivariate Correlation Plot:

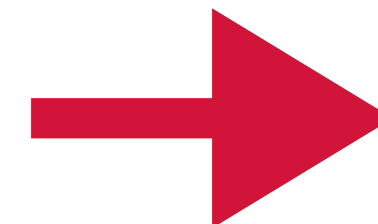
	Cholesterol	Calories	Fat	Fiber	Female	Male	Age	Alcohol	Smoker/NonSmoker	Vitamin Use
Cholesterol	1									
Calories	0.6592	1								
Fat	0.7098	0.8718	1							
Fiber	0.1540	0.4655	0.2765	1						
Female	-0.2550	-0.2074	-0.1958	-0.0461	1					
Male	0.2550	0.2074	0.1958	0.0461	-1	1				
Age	-0.1136	-0.1768	-0.1695	0.0449	-0.2804	0.2804	1			
Alcohol	0.1823	0.4515	0.1857	-0.0201	-0.2278	0.2278	0.0516	1		
Smoker/NonSmoker	0.0907	0.0769	0.0857	-0.1639	-0.0345	0.0345	-0.1533	0.1409	1	
Vitamin Use	-0.0326	0.0089	-0.0335	0.1034	0.1355	-0.1355	0.0123	-0.1209	-0.1345	1



Model Assumptions:

- Residuals are linear
- There is no multicollinearity
- There is no autocorrelation
- Errors are normally distributed

- The dependent variable Cholesterol is highly correlated with Calories and Fat followed by a relatively low correlations with rest of the factors.
- The independent variables Fat, Calories and Alcohol are correlated among each other indicating there could be multicollinearity. A VIF of 5 and below is considered as a cutoff for the model.



Testing Individual & Joint Significance

Testing the individual significance of each independent variable, prioritizing variables with high correlations moving towards lower ones.

Individual Significance and Coefficients:

Variable	Coefficients	P-Value
Fat	2.7696	0.00000
Calories	0.1279	0.00000
Fiber	3.8127	0.00618
Age	-1.0288	0.04392
Alcohol	1.9522	0.00116
Female	-98.8421	0.00000
Male	98.8421	0.00000
Vitamin Use	-5.0008	0.56420
Smoker/NonSmoker	34.8259	0.10800

Results:

- The P-values indicate that fat, calories, fiber, alcohol, age, and gender significantly impact cholesterol levels.
- Vitamin Use and smoking habits are not significant in explaining cholesterol levels.

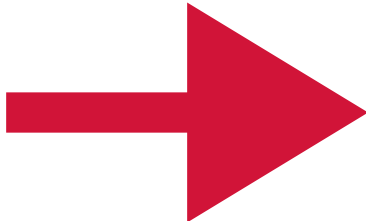
Testing the joint significance by prioritizing variables with high correlations that were individually significant.

Goodness of fit statistics for the joint model:

Source	Value	Standard error	t	Pr > t	Lower bound (95%)	Upper bound (95%)
Intercept	32.9338	16.1134	2.0439	0.0418	1.2284	64.6393
Calories	0.0485	0.0176	2.7550	0.0062	0.0139	0.0831
Fat	1.9493	0.3246	6.0051	0.0000	1.3106	2.5879
Female	42.7259	15.4383	2.7675	0.0060	12.3488	73.1030
Fiber	-2.6141	1.1415	-2.2901	0.0227	-4.8601	-0.3681

Observations	315
Sum of weights	315
DF	310
R ²	0.5312
Adjusted R ²	0.5252
MSE	8272.6339
RMSE	90.9540
MAPE	29.4618
DW	2.1253
Cp	5
AIC	2846.4830
SBC	2865.2458
PC	0.4839

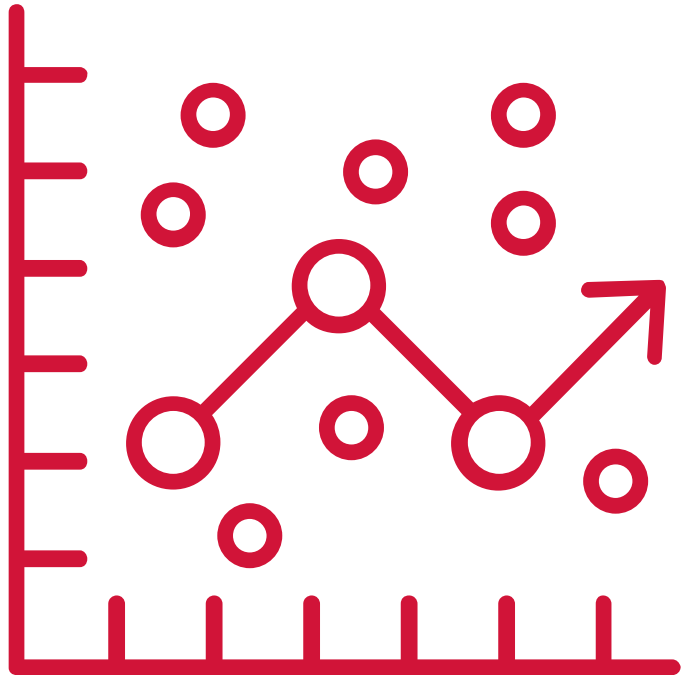
Level of Significance:
5 % level of significance.



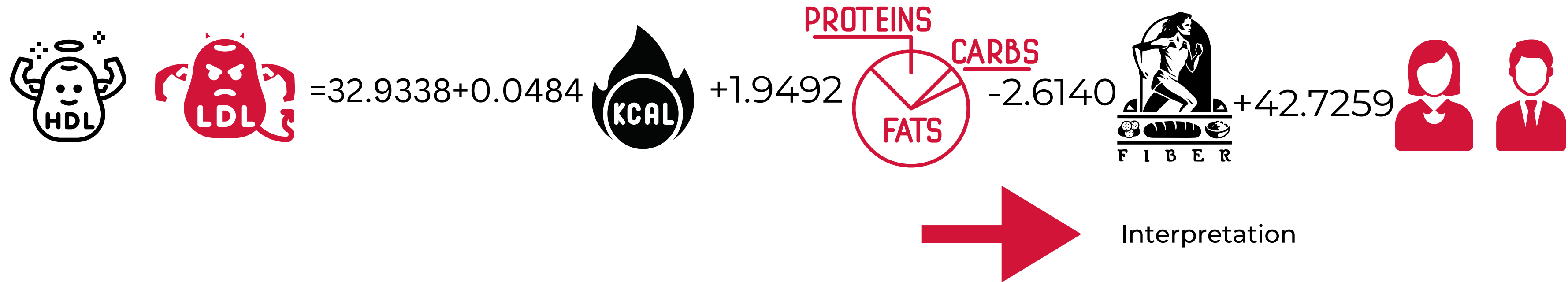
Model Parameters

Model Parameters:

Source	Value	Standard error	t	Pr > t	Lower bound (95%)	Upper bound (95%)
Intercept	32.9338	16.1134	2.0439	0.0418	1.2284	64.6393
Calories	0.0485	0.0176	2.7550	0.0062	0.0139	0.0831
Fat	1.9493	0.3246	6.0051	0.0000	1.3106	2.5879
Female	42.7259	15.4383	2.7675	0.0060	12.3488	73.1030
Fiber	-2.6141	1.1415	-2.2901	0.0227	-4.8601	-0.3681



Model: Cholesterol = 32.9338+0.0484*Calories+1.9492*Fat-2.6140*Fiber+42.7259*Female Dummy

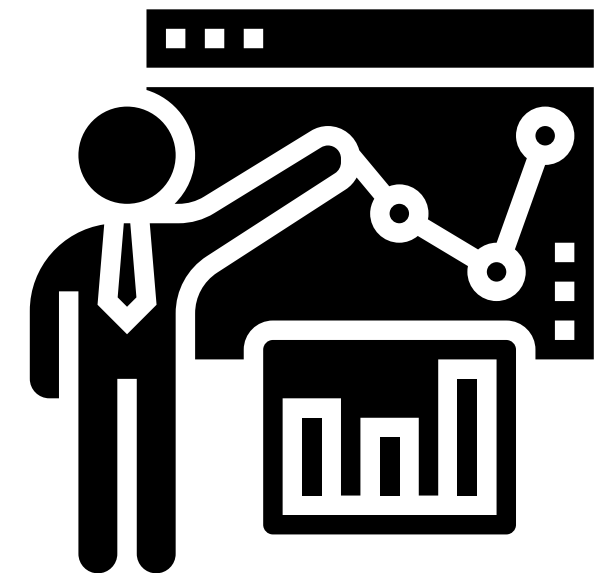


Interpretation:

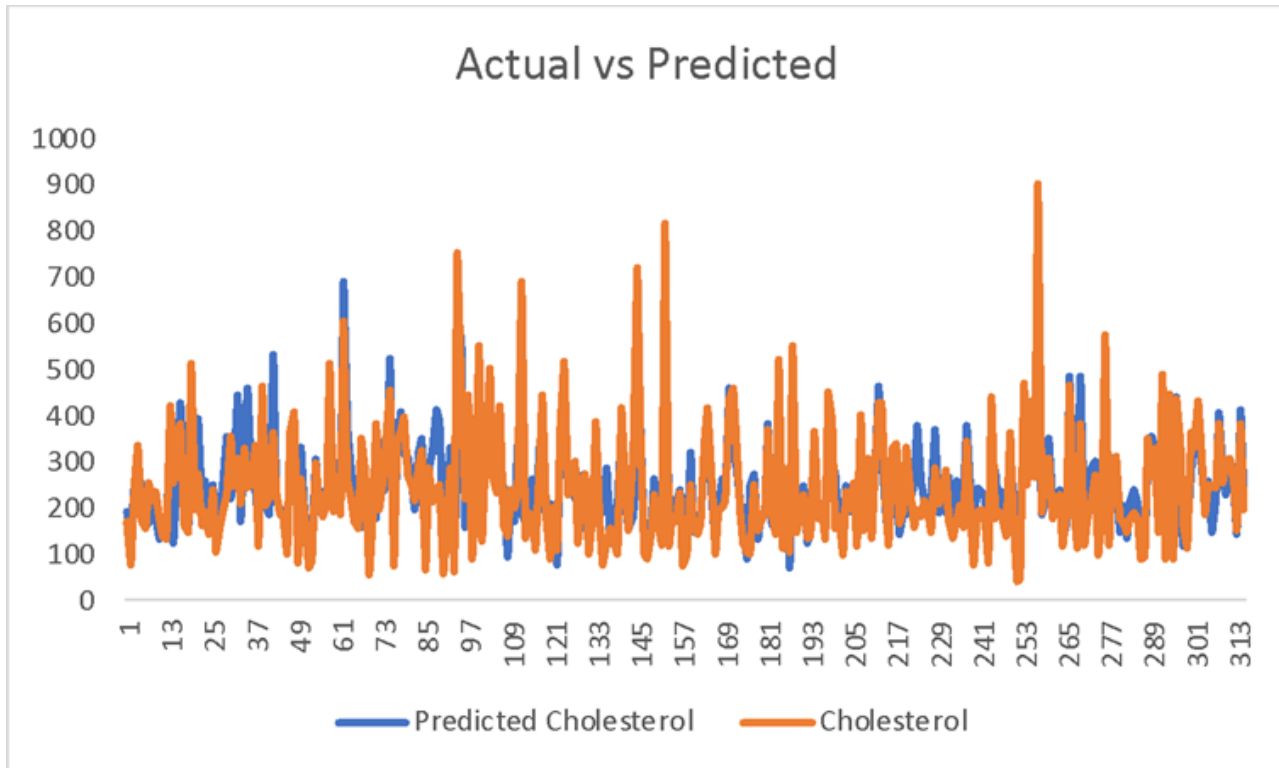
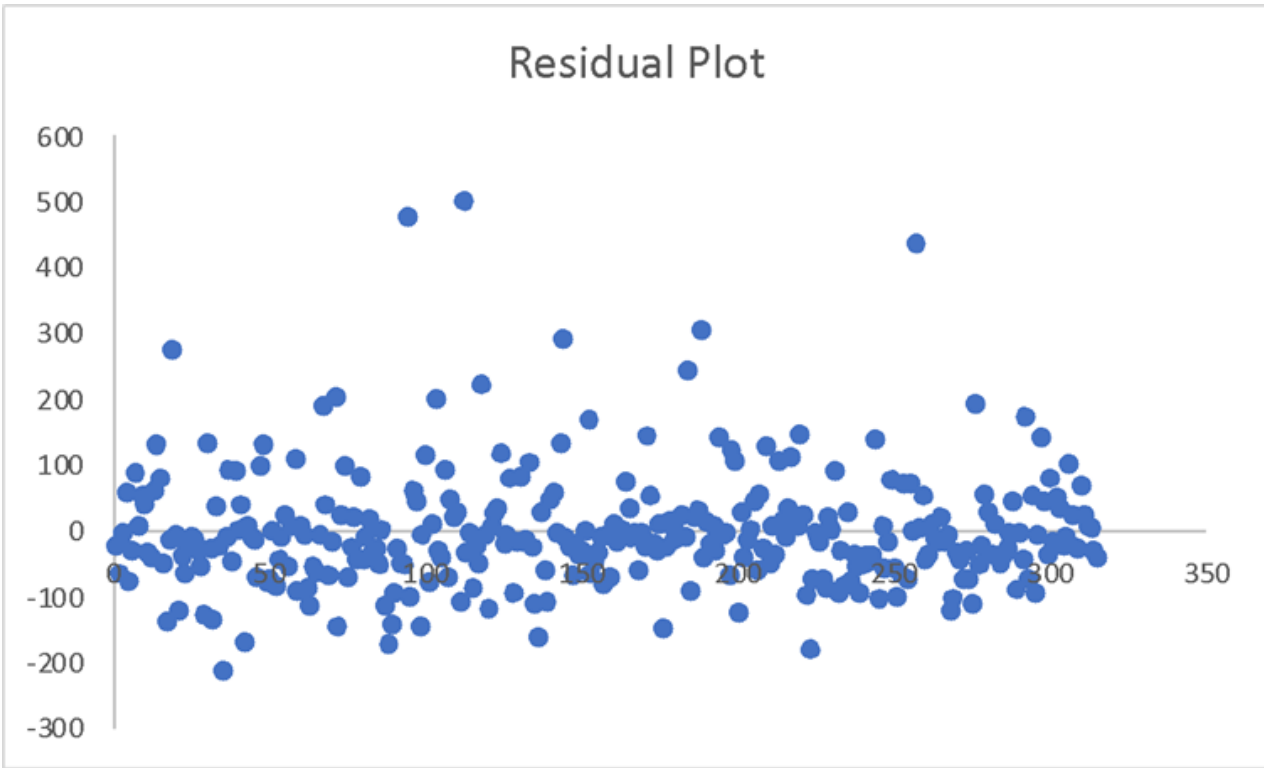
- Fat, Calories, Fiber, and Gender jointly impact cholesterol, with fat being highly significant, followed by gender, calories, and fiber.

The model coefficients indicate:

- One-milligram increase in fat contributes to 1.94 milligrams of increase in cholesterol.
- One-kilo increase in calories contributes to 0.04 milligrams of increase in cholesterol.
- Gender is associated with high levels of cholesterol.
- One gram increase in fiber leads to 2.61 milligram of decrease in cholesterol.

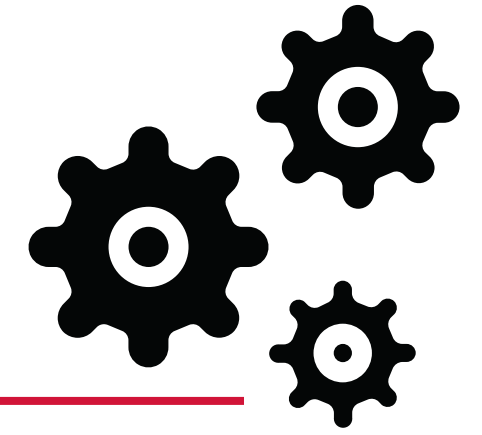


53% of variation is explained by independent variables in explaining cholesterol



Errors are normal with a few outliers. The outliers seemed to be normal across variables and did not seem like a data entry error, these extreme observations looked like the data from patients who were overweight. However, we do not have weight related data available to relate to the extreme observations. This regression model is built without treating for outliers. Outliers can be influential observations.

Binary Logistic Regression Analysis



Area of interest

To find the probability that a patient is going to be a cholesterol high risk given other factors like Calories, Fat, Gender, Age, Alcohol, Smoker/Nonsmokers and Vitamin Use.

- General guidelines to interpret cholesterol test results

Cholesterol Level	
Below 200 mg/dL	Desirable
200-239 mg/dL	Borderline high
240 mg/dL and above	High

- For the Binary Logistic Regression Model, we considered these cutoffs. Observations for which cholesterol levels were below 200 were categorized as 0(Low Risk) and cholesterol levels of over 200mg were categorized as 1(High Risk).

Cholesterol Level	Risk	No of Patients
Below 200mg	Low risk	148
Over 200mg	High Risk	167

Hypothesis:

Null Hypothesis: All the coefficients of the regression model are equal to zero.
None of the independent variables are significant in explaining the probabilities.

Alternative Hypothesis: At least one of the variables is significant in explaining the probabilities.

The Predicted probability model:

Model Parameters:

Source	Value	Standard error	Wald Chi-Square	Pr > Chi²
Intercept	-6.1246	0.9091	45.3871	0.0000
Calories	0.0040	0.0005	65.8359	0.0000
Fiber	-0.1491	0.0367	16.4910	0.0000
Age	0.0260	0.0103	6.3023	0.0121

Statistic	Independent	Full
Observations	315	315
Sum of weights	315.000	315.000
DF	314	311
-2 Log(Likelihood)	435.536	297.691
R²(McFadden)	0.000	0.316
R²(Cox and Snell)	0.000	0.354
R²(Nagelkerke)	0.000	0.470
AIC	437.536	305.691
SBC	441.289	320.701
Iterations	0	5

$$\text{Pr (Chol risk=1)} = 1 / (1 + \exp(-(-6.124600+0.004001*\text{Calories}-0.149103*\text{Fiber}+0.025981*\text{Age})))$$



Inference:

- The regression coefficients for calories and age being positive indicate a positive influence on the probabilities.
- The regression coefficients of fiber are negative, indicating a negative influence on the probabilities.

Classification table:

from \ to	0	1	Total	% correct
0	114	34	148	0.7703
1	36	131	167	0.7844
Total	150	165	315	0.7778

Out of 167 observations related to high cholesterol levels 78.44% (131 records) were classified correctly.

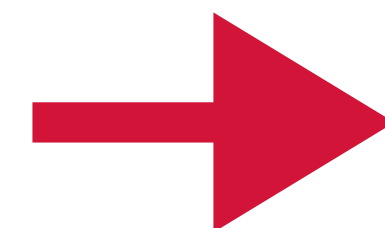
Changing Impact, an illustration:

Calories, Fibre and Age	Probability of Cholesterol High Risk
When calories=1000, fibre=4 and age=20	0.0997
When calories=2000, fibre=4 and age=20	0.8582
When calories=2000, fibre=20 and age=40	0.4837
When calories=2000, fibre=20 and age=50	0.5484

	Calories	Fiber	Age
Minimum	445.2	3.1	19
Maximum	6662.2	36.8	83
Average	1796.7	12.8	50

Comparison of first two cases: Probability of risk increases by 0.7585 when calorie consumption increases by twice the amount with same age and same fiber intake.

Comparison of Last two cases: On the other hand, with increase in age probability of risk increases by only 0.0648 when consuming same amount of calories and fiber.



Conclusion

Conclusion

Based on our data, we found the following

- The means for calories, age, fat, and cholesterol differ between males and females.
- The means for age, fiber, and vitamin use differ between the smoker and non-smoker groups.
- Smoking changes with age; for instance, people over 50 smoke more than those under 35, but those between 35-50 smoke the least.
- Alcohol consumption changes with age; for instance, people over 50 drink less than those between 35-50, and those under 35 drink the least.
- Vitamin use is consumed mostly by people between 35-50, and people over 50 come after and least by people under 35.
- The means for fiber and calories differ in terms of age.
- The increase in calories and fat increases cholesterol levels; on the other hand, an increase in fiber decreases cholesterol levels. Nevertheless, age and gender are also associated with cholesterol levels.

Thank You

Prevention is
Better than Cure